



Feasibility as a Parameter

When comparative research on narrow specialist populations can and cannot be executed

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THE EVIDENCE IS MY OWN FAILED STUDY

334,976

panel members screened

→ 20 eligible

→ 4 usable

Six per hundred thousand. I did not set out to study this. The constraint arrived as a result.



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ELEMENT 1 · OVERALL RESEARCH TOPIC, AND WHY IT MATTERS

Comparative research has a requirement it never states as one

To compare a phenomenon across countries you must reach the same population in every country, at the same time, with an instrument that means the same thing in each place.

WHEN THE POPULATION IS GENERAL

The requirement is demanding but routine. Effort and money convert into responses at a rate that differs by country and is always positive, always improvable.

A FIELDWORK PROBLEM

WHEN THE POPULATION IS RARE

Conversion is governed by arithmetic fixed before the first contact. That arithmetic does not add across borders. It compounds, and in one direction only.

A DESIGN PARAMETER

A design that is marginal in one country is not four times harder in four. It can be structurally impossible in four while remaining marginal in each one separately.



1 ELEMENT 1 · OVERALL RESEARCH TOPIC, AND WHY IT MATTERS

The gap, and the question

THE TOPIC

Reaching a narrow specialist professional population in several national frames at once.

THE RECEIVED WISDOM

Sampling difficulty is an execution problem. It belongs to budget and local partners, and it appears in print as a limitation paragraph.

THE GAP

Nobody has asked what happens when the population is rare. The quantities that decide it are observable before fielding, and they compound across frames.

THE RESEARCH QUESTION

Under what conditions is a comparative multi-country design on a narrow specialist population feasible at all, and where must a survey give way to another method?

THE CONTRIBUTION

To move sampling feasibility from the status of a LIMITATION to the status of a PARAMETER. A parameter is estimated before a design and constrains it. A limitation is confessed after the design has failed.

WHO BENEFITS

Researchers designing comparative work · doctoral committees assessing feasibility · buyers of multi-country research



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ELEMENT 2 · THE STUDY MODEL

The model: four terms, four different levels

Reachable usable sample is a product, not a sum. Each term is decided at a different level, and attributing a failed frame to the country assigns to one level what another produced.

LEVEL OF ANALYSIS

One occupation, on one provider, in one national frame. Cells nested in national frames, frames nested in regions, providers cross-classified.

FRAME SIZE

panel members

POPULATION

× PREVALENCE

how rare the specialty is

INFRASTRUCTURE

× RESPONSE RATE

who answers

INSTRUMENT

× SCREEN SURVIVAL

who survives eligibility

DESIGN



= REACHABLE USABLE SAMPLE

observable before a study is fielded, at no cost, on the panel's own interface



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ELEMENT 3 · PRELIMINARY HYPOTHESES, AND THEIR JUSTIFICATION

Five hypotheses

EACH TAKES ITS LOGIC FROM ONE OF THIS WEEK'S FOUR PAPERS, CITED IN THE LINE ITSELF

SIGN		
H1	≤ NEVER RISES	Joint feasibility is set by the least feasible frame, not the mean. Adding a country leaves it unchanged or lowers it, and can never raise it. Analytic: the structure of a conjunction.
H2	+ POSITIVE	Following Rugman and Verbeke (2004) on home-region orientation, specialist prevalence on a panel is higher inside the panel provider's home region.
H3	+ POSITIVE	Following Lopez, Kundu and Ciravegna (2009) on born-regional firms, studies described as cross-national achieve regionally concentrated coverage, more so as the population narrows.
H4	> GREATER	Following Meyer, Li and Schotter (2020) on levels of analysis, mode explains more variance in response than country does.
H5	— NEGATIVE	Extending Zeng, Groggaard and Bjorkman (2023) on control against coordination, standardization is negatively associated with response, and worst in the frames that already bind.

THE CONTRIBUTION IS H2

H1 is the structural result the design rests on. H2 carries the international business contribution, applying the regionalization thesis to research infrastructure.

UNSIGNED ON PURPOSE

H1 is analytic, so there is no coefficient to estimate. H4 compares variance components rather than predicting a direction.



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ELEMENT 3 · PRELIMINARY HYPOTHESES, AND THEIR JUSTIFICATION

H1 · A comparison is a conjunction, not an average

To say a relationship differs between Spain and China is to assert something about Spain and something about China. The assertion fails if either component fails.

AN AVERAGE CANNOT SATISFY A CONJUNCTION

A design with one excellent frame and one hopeless frame has a respectable mean and no comparison. The requirement is set by the minimum.

Because a set's minimum can only fall or stay level when an element is added, every additional country weakly worsens the design and no additional country can improve it.

Breadth is treated in comparative work as a virtue. It is purchased at a price that rises faster than the count of frames. A design covering more countries has not become more ambitious. It has become more fragile.



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ELEMENT 3 · PRELIMINARY HYPOTHESES, AND THEIR JUSTIFICATION

H2 and H3 · A research panel is a firm

Of the 380 largest multinationals with usable sales data, Rugman and Verbeke found 320 home-region oriented, averaging 80.3 percent of sales at home. Nine were global. Their mechanism is specific: upstream advantages in technology travel, downstream advantages such as branding are location-bound.

A research panel is a firm whose entire product is a downstream asset: its membership, and its standing with that membership. Survey software is a commodity, so there is almost no upstream advantage to carry abroad. On their own account it should be MORE home-region bound than a manufacturer, not less.



WHY IT BITES HARDER FOR SPECIALISTS

A consumer panel is built by broad advertising anywhere. A panel of experienced auditors is built through professional bodies whose reach is national by construction.

AND WHY THE SAMPLE COLLAPSES

Of 40 Costa Rican software firms, exactly one had exported from inception to the market the founders called strategic. The rest sold to neighbours. The category could not see the difference.



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ELEMENT 3 · PRELIMINARY HYPOTHESES, AND THEIR JUSTIFICATION

H4 and H5 · Level of analysis, and the price of equivalence

H4 · MODE OVER COUNTRY

Response is the outcome of an encounter between a person and a request. Whether it arrives by telephone from a human being or by email from a platform is a property of the instrument, not of the country.

Mode changes the cost of refusing. Country changes only the disposition toward it.

THE ONE DATUM

Harzing's 47% in Korea was telephone. It exceeds the entire country spread in the same project by about five times. One observation is not evidence. It is why this is worth testing.

H5 · CONTROL AGAINST COORDINATION

Zeng and colleagues name standardization as one of four organizational mechanisms, on the control side, against coordination by mutual adjustment. They do NOT claim the two trade off. They say the interaction is understudied.

Equivalence needs an identical instrument. Response needs adaptation of length, channel, sponsorship and register. Proposing that these pull against each other is my conjecture about a gap they name, not a finding I am borrowing.

A strong frame can afford the loss. A weak frame had no margin, and under H1 the weak frames bind the design.



4 ELEMENT 4 · HOW I WOULD STUDY IT EMPIRICALLY · DATA SOURCES

What gets measured, and why it can be measured at all

Panel providers expose an audience-configuration interface that returns an estimated reachable count once screening criteria are entered, before any commitment and at no cost. That is what makes feasibility a parameter rather than a limitation.

LEVEL OF ANALYSIS	Three levels. The cell, the national frame, the region. Providers cross-classified
THE CELL	One occupation, on one panel provider, in one national frame
THE MATRIX	6+ occupations × 8+ national frames × 3+ providers ≈ 144 cells
REGION IS A VARIABLE, NOT A SETTING	Providers chosen to vary on home region. With one region there is no variation and H2 is unidentifiable
THE VALIDATING CASE	My own study. A model that cannot reproduce 334,976 → 20 → 4 is rejected

NO DATA HAVE BEEN COLLECTED FOR THIS PROJECT. THIS IS THE DESIGN, NOT A RESULT.



What gets measured, and how

DEPENDENT VARIABLE

Reachable usable sample = frame size x prevalence x response rate x screen survival

Source: the panel audience-configuration interface, read free before any commitment. Second dependent variable for H4 and H5: achieved response rate, from published method sections

INDEPENDENT VARIABLES

	VARIABLE	HOW MEASURED	WHERE THE DATA COMES FROM
H2	Region concordance	Binary, triad classification	Provider incorporation and headquarters filings
H3	Coverage discrepancy	Stated minus achieved regional concentration	Method sections of published cross-national studies
H4	Mode	Categorical: phone, email, panel, in person	Method sections, coded by two readers
H5	Standardization index	Wording, length, incentive, sponsorship, protocol	Published instruments and appendices

CONTROLS Prevalence · frame size · eligibility strictness, as the count of conjunctive screening criteria · incentive at purchasing power parity (PPP) · field period · year



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ELEMENT 4 · HOW I WOULD STUDY IT EMPIRICALLY · METHOD

Two different kinds of claim, two different treatments

H1 · ANALYTIC

Established by demonstration and probed by sensitivity analysis across the plausible range of prevalence, response and screen survival, reporting the region of the parameter space where a comparative design stays viable.

The claim is about the structure of a conjunction. It would not be made more true by a p-value, and reporting one would present a deductive result as an empirical one.

H2 – H5 · INFERENCE

Hierarchical linear models. Cells nested in national frames, frames nested in regions, random intercepts at frame and provider level.

The nesting is not a convenience. It is the structure Meyer and colleagues argue must be modelled, and ignoring it attributes to countries the variance that belongs to providers.

H5 IS SUPPORTED ONLY BY THE INTERACTION, NOT THE MAIN EFFECT

SOFTWARE

R, with lme4 for the hierarchical models and base R for the sensitivity analysis.

Stata as a cross-check on the multilevel estimates. All code and data versioned in git.



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ELEMENT 4 · HOW I WOULD STUDY IT EMPIRICALLY · TIMEFRAME

Nine months, and no recruitment step



The constraint that ended my qualifying study was recruitment. This design has no recruitment step at all. The measurements already exist on the platforms, so the work is gathering them systematically rather than obtaining them.



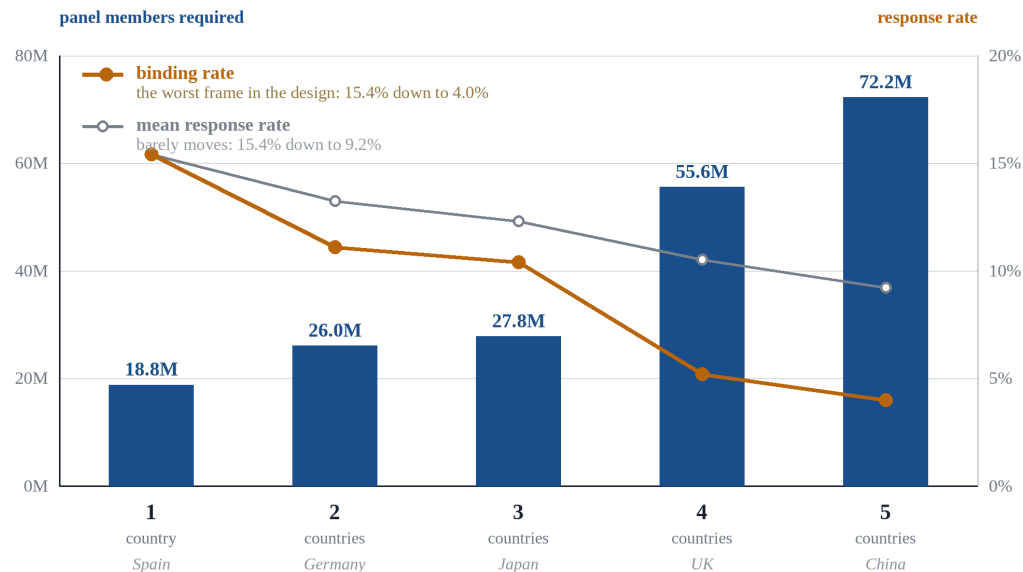
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ELEMENT 5 · PRELIMINARY ANALYSIS

Preliminary analysis: the design is priced by its worst frame

Adding a country prices the design by its worst frame, not its average one

Panel members required in EVERY national frame to yield 30 usable responses, at a prevalence of 6 per 100,000



Response rates: Harzing, Reiche & Pudelko (2012), Illustration 7.

Prevalence and screen survival: the author's own study (20 eligible of 334,976; 4 usable of 23 raw).

ADD FOUR COUNTRIES

Mean response rate

15.4% → 9.2%

barely moves

Panel needed in EACH

18.8M → 72.2M

a factor of four

The design is priced by its worst frame, not its average one. Reporting an overall response rate reports the wrong statistic.



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ELEMENT 5 · PRELIMINARY EVIDENCE

The evidence I have is my own failed study

No data has been collected for this project. What follows is from the qualifying study, which was designed to test something else and failed for the reason this project is about.

WHAT HAPPENED

334,976

panel members screened against the eligibility criteria

→ ~20 eligible

→ 4 usable, of 23 raw starts

ABOUT SIX PER HUNDRED THOUSAND

WHICH HYPOTHESES IT BEARS ON

H1

Directly. It supplies the prevalence and screen-survival terms the whole model runs on

H2

Not yet. One provider, one home region. The comparison H2 needs does not exist here

H4

Weakly. My instrument was one mode, so mode could not be separated from anything

This is one case, fully documented, and it is the reason the project exists. It is not evidence that the pattern generalises, which is what the design above is for.



What this model does not claim

Prevalence is held constant across countries

It almost certainly is not. No cross-national prevalence data for these populations exists, which is why obtaining it is the contribution rather than an assumption.

The response rates come from one study

One instrument, one period. Treating them as a general country characteristic would commit the level-of-analysis error the paper itself identifies.

Rates are treated as independent

They are not. Frames within a region share institutional conditions and will correlate.

Panel estimates are vendor-supplied and unaudited

They are the same numbers researchers already rely on, so this adds no new exposure, but estimated against achieved is a validation step not yet in the design.



THE CONTRIBUTION

A parameter is estimated before a design. A limitation is confessed after it fails.

Comparative international business has treated who can be reached as a matter of execution. For general populations that is harmless. For narrow specialist populations it is not, because the arithmetic compounds across frames in one direction and is decided by the weakest frame rather than the average one.

The quantities are observable before a study is fielded. Making them part of the design, rather than part of the confession, is a small change in practice and a substantial change in what the field can claim to have compared.

AND THE SECOND IMPLICATION

Where the arithmetic rules out a survey it does not rule out the study. Twenty reachable participants are a failed survey and a well-powered phenomenological one.

THE THRESHOLD IS LOCATABLE IN ADVANCE